Chapter 15

# Getting the appearance correct

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## CHAPTER CONTENTS

**Introduction** 187
**Light** 187
**Human anatomy and colour vision** 189
Colour vision deficiency (colour blindness) 189
**Description of colour** 189
Munsell's colour order system 189
Hue 189
Chroma 190
Value 190
**Tooth shade selection** 190
Traditional shade matching 190
Shade guides for indirect restorations 190
Effects of colour vision deficiencies 192
Instrumental shade matching 192
Shade-matching devices 192
Digital photography 192
**Supplemental aesthetic information** 194
Aesthetic analysis 194
Facial relationships 194
Dento-facial relationships 195
Dento-gingival relationships 196
**Summary** 199

## INTRODUCTION

When making restorations that are visible in the smile line, and for some patients even when the restorations are not readily visible, appearance is critical. Creating a good appearance in a restoration depends upon matching the colour, shape and surface texture to adjacent teeth. Whilst selecting the correct shade requires an understanding of light and the science of colour, creating the shape and texture of teeth needs knowledge of tooth morphology, dimensions and proportion. This chapter addresses these issues so that the optimum appearance of aesthetic restorations can be achieved.

## LIGHT

Visible light forms a small portion of the whole electromagnetic (EM) spectrum (Figure 15.1) and often takes the form of polychromatic light which is composed of electromagnetic radiation of more than one wavelength. The colour of an object that one observes is actually the reflection of the light that strikes it. For example, a red flower appears red because red light is reflected by the flower whilst the other colours of light are absorbed.

The unique appearance of teeth is due to the complex interactions between light and tooth tissue, and this makes shade selection difficult. The interactions that take place are as follows:

- *Reflection*. Light is reflected by mineralized tissue, mainly enamel rods and extratubular dentine. The amount of reflected light helps to determine the brightness of a tooth: the more light that is reflected, the brighter the tooth will appear.
- *Scatter*. Light entering the tooth tissue hits various tooth structures and is dispersed in all directions. Some light is returned towards the source and emerges from the tooth surface again in all directions. Scattering therefore reduces the intensity of the radiation being returned from the material. What

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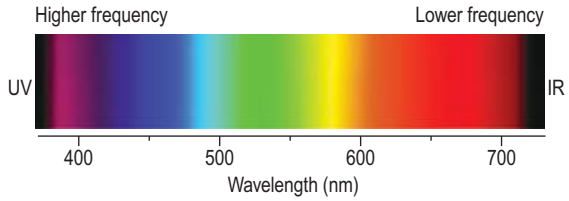



**Figure 15.1** Electromagnetic spectrum.

causes the majority of scatter is different in each dental tissue. For example, in enamel a large amount of scatter is due to the enamel prisms, specifically the hydroxyapatite crystallites. However, most of the scatter in a tooth occurs in dentine and this is due to the dentinal tubules.

- *Absorption*. Light is absorbed by teeth and occurs to a greater extent in dentine than enamel. Energy that is absorbed does not emerge from the material and is converted to another form of energy. The amount of absorption is dependent on the light source and tooth.
- *Refraction*. The change in direction of a beam of electromagnetic radiation due to a change in the conveying medium is termed refraction. This occurs to a small extent in enamel (97% mineralized by weight) and to a greater extent in the less mineralized and tubular structure of dentine (approximately 70% mineralized by weight)
- *Transmission*. Light transmission occurs in both enamel and to a lesser extent in dentine. This is evident as neither material is opaque, unlike some materials used in dentistry (Figure 15.2). Transmitted light radiation passes through the incisal edge and approximal areas of a tooth following a number of the reactions described above. The number of these reactions is reduced at the incisal edge compared to the body of a tooth. This results in more light being transmitted at the incisal edge, occasionally giving the appearance of an almost transparent region. Enamel is almost translucent and if no dentine was present to block the transmission of light, teeth would appear glass-like.



**Figure 15.2** Opaqueness of the metal–ceramic crown on the upper left lateral incisor compared to the more translucent teeth and resin composite restorations.



**Figure 15.3** Image of teeth that have dehydrated under a rubber dam; the arrow shows where the rubber dam sat.

Dentine, however, provides the colour of a tooth that is evident in the body and cervical areas. When a tooth is dehydrated, it appears whiter and brighter due to dehydration of the collagen matrices. This can occur after impression making, rubber dam placement (Figure 15.3) and bleaching. It is therefore important to shade match at the beginning of an appointment rather than at the end when some temporary colour change might have occurred.

- *Translucency*. Translucency is defined as the ability to allow radiation to pass with little scatter. The degree of translucency varies with the wavelength of the light. Enamel is translucent due to the high level of mineralization and crystallite orientation. However, when enamel is hypomineralized as a result of developmental (fluorosis) or acquired (carious white spots – see Chapter 1 on caries) defects, greater scattering of light occurs and the enamel becomes opaque and less translucent.
- *Fluorescence*. Fluorescence is the property of a material to absorb light of a particular wavelength and then to emit light of a different wavelength and therefore colour. Teeth fluoresce bluish-white when exposed to ultraviolet radiation.
- *Metamerism*. Metamerism can have an effect on the appearance of a tooth or restoration. Metamers are



**Figure 15.4** Metamerism: The same shade tab and tooth are matched under different light sources. It is observed that the shade match is better when viewed under the centre picture.

objects which match each other under one set of light conditions but mismatch under another as the light has different spectral properties (Figure 15.4).

- *Opalescence*. Opalescence is defined as having the property of opal stone. Natural teeth have the same properties as these stones: they appear yellowish-red in transmitted light but blue in reflected light.

## HUMAN ANATOMY AND COLOUR VISION

The retina, on the internal posterior wall of the human eye, contains a complex network of nerve endings capable of detecting light. There are over 120 million light-sensitive receptors in the 0.2 mm thick retina. Two types of receptor cell are present: rods and cones. Rods far outnumber cones and are responsible for night vision which is monochromatic. Cones are responsible for medium to high level light vision in full colour and are found in the centre of the retina. Colour judgement is theoretically impaired if the operator views an object from the side of the eye. Three types of cone exist; each one has a different sensitivity to different wavelengths of light (blue, red and green).

### Colour vision deficiency (colour blindness)

Colour blindness is usually a sex-linked inherited condition but is rarely due to an acquired defect of the retina; as a result, the majority of people who suffer from this condition are male. Approximately 8–10% of the male population suffers from a congenital colour vision defect (CVD) whilst the prevalence among females is much less (approximately 1%). Differences in prevalence of CVD have also been found between different racial groups. The affects of a CVD on shade selection is discussed below.

## DESCRIPTION OF COLOUR

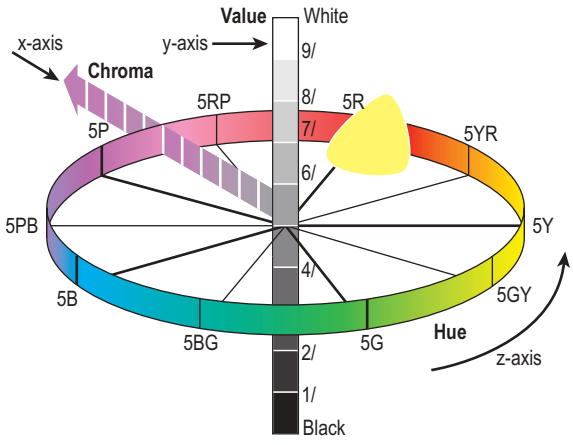
There are three main systems which can be used to describe and quantify colour, namely Munsell's colour order system, the 1976 C.I.E.L*a*b* uniform colour space system (C.I.E Commission Internationale d'Eclairage (International Commission on Illumination)) or the RGB (Red, Green, Blue) colour space system. The first system is the one most commonly used and quoted in clinical dentistry and is therefore the only system described in this text.

### Munsell's colour order system

The Munsell colour system was devised by a painter, Albert Munsell, in 1905. The system's attributes are hue, chroma and value (Figure 15.5).

#### Hue

Hue is the quality by which we distinguish one colour from another – for example, red from yellow, or green from blue. There are 100 Munsell hues: 10 major hues with each placed 10 steps apart in a horizontal plane (z axis, Figure 15.5) around a central axis. Teeth are found in the yellow and yellow–red region but the exact range of hues varies with the method of assessment and is different for extracted teeth. Dentine provides the main source of hue in a tooth but this is modified by the enamel.




Code : R=Red, Y=Yellow, G=Green, B=Blue, P=Purple

**Figure 15.5** Framework of Munsell's colour order system.

### Chroma

Chroma describes the intensity of the colour (hue) and distinguishes a strong colour from a weak one. The chroma scale starts at /0 and extends outwards; its maximum varies with each hue. The purest colours are found at the extremes of the colour cylinder (x axis, Figure 15.5). Reduced thickness or mineralization of dentine usually results in decreased chroma. The chroma of teeth is usually found within the range /1 to /5 but can range from /0 to /7.

### Value

Value is the quality by which one distinguishes a light colour from a dark one and is therefore the brightness on a grey scale. The value symbol 0/ is used for absolute black and 10/ for absolute white (y axis, Figure 15.5). In a healthy young tooth there is less dentine thickness due to a reduced amount of secondary dentine and so the ratio of reflected to absorbed light radiation is increased compared to older teeth: as a tooth ages its value therefore decreases. Tooth value is usually found within the range 6/ to 8.5/ but can range upwards from 4/.

## TOOTH SHADE SELECTION

'Selecting the shade' for a restoration belittles the complexity of the process of determining the shade and form for a restoration. Providing the technician with only the correct shade will not enable them to fabricate an aesthetic restoration, a substantial amount of additional information is required. Additionally, the clinician has to prepare teeth in such a way that the technician is able to recreate the desired shape and shade. (See chapter 10, 11 and 12). Methods of traditional and instrumental shade-matching will be described in this chapter, together with the supplemental information required for an aesthetic restoration.

## Traditional shade matching

The basic shade for a restoration is usually selected using a shade guide. Most manufacturers provide a shade guide for use with their materials, whether they are for use with indirect or direct restorations. These allow transfer of information relating to tooth shade from the clinic to the laboratory. Ideally, shade selection would be completed by both the dentist and technician in conjunction with the patient. Unfortunately, this is often not practical as the laboratory is usually some distance from the dental clinic.

Value (brightness) is regarded as the most important and discernible aspect of colour when selecting a shade. Metal–ceramic crowns (MCC) have a higher value than natural teeth and are less translucent. The opaque ceramics that are required to mask the metal coping, which is obviously not translucent, give them their high value (see Chapter 10) and expose the restoration as fake compared to natural teeth (Figure 15.6). The increased value of crowns is usually due to insufficient tooth reduction (for adequate dentine and enamel ceramic) and laboratory shortcomings. A higher value also makes the tooth/restoration appear larger.

### Shade guides for indirect restorations

The most commonly used shade guide for indirect restorations is based on the VITA classical shades. The VITA classical shade guide can be organized in two ways. First, the tabs can be grouped by similar hues (A–D), with these groups being divided via numerical values (1–4). Generally, the chroma (intensity of colour) increases and value (lightness) decreases as the numbers rise (Figure 15.7, above). Second, the shade tabs can be arranged by decreasing value (Figure 15.7, below).

The VITA System 3D-Master shade guide (Figure 15.8) arranges the shade tabs in a more logical fashion and is currently the only shade guide to meet the relevant American Dental Association (ADA) standard. The tabs are divided into five value groups (six if the three bleached shades are included). As value decreases, the tab group number increases. Within each value group, the chroma increases as one descends down the group. Moving from right to left alters the hue, making the tabs more red and yellow, respectively.



**Figure 15.6** The effect of value on appearance. Metal–ceramic crowns on the central incisors (left) have too high a value and look bright. Replacement crowns (right) after correct tooth reduction and laboratory procedures give a more pleasing appearance.




**Figure 15.7** VITA classical shade guide arranged by hue (above) and value (middle and below).

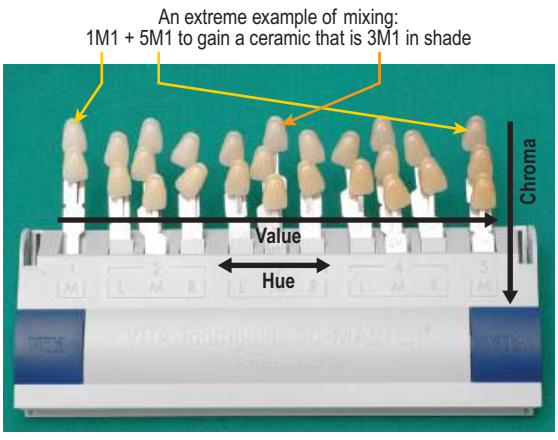



**Figure 15.8** VITA 3D shade guide.

Unlike the VITA classical shade guide and most other shade guides, the manufacturer claims that each tab is equally spaced from the other, making intermediate shade selections possible. For example, 50% 1M1 plus 50% 5M1 will produce a 3M1 ceramic, or more relevant to the clinical setting, 75% 1M1 plus 25% 1M2 will produce a 1M1.25 ceramic. This allows a more objective selection of intermediate shades, making up to 96 different combinations possible. The fact that the tabs are equally spaced has been questioned, but despite this it has improved shade matching.

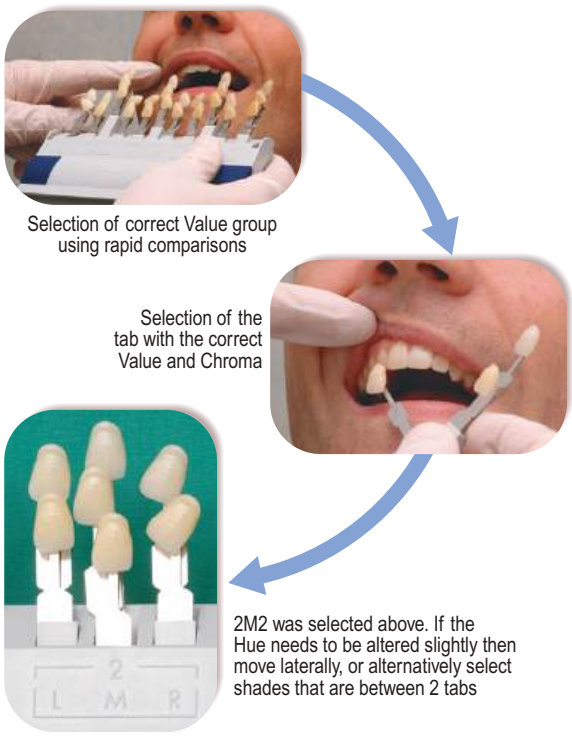



**Figure 15.9** Shade selection using VITA 3D Master System.

The manufacturer's instructions should be used when selecting a shade with this shade guide (Figure 15.9). If the VITA System 3D-Master is used, it is important to ensure that the technician has the corresponding ceramic: if a 3D shade is selected, the restoration cannot be fabricated from VITA classical porcelains. Whilst a conversion table is available to convert VITA classical into 3D shades, this is only approximate.

Shade matching for teeth is notoriously difficult and is a result of a combination of problems:

- The unique properties of the teeth discussed previously
- Inconsistency in material manufacture
- Human observer errors.

There are many materials used to create aesthetic restorations; however, these do not always exhibit colour consistency. There are visible colour differences between batches and brands of ceramic and composite resin. Additionally, shade guides do not match each other, correspond to the colour of teeth or always match the materials they represent. Figure 15.10 shows three different A3 materials compared to a reference VITA classical shade tab and the VITA 3D shade tab that approximately represents A3. Custom shade guides manufactured from the batch of ceramic and composite resin to be used have been suggested, but these are not a practical alternative.




**Figure 15.10** Two shade tabs and three materials all purporting to have an A3 shade. Obvious differences can be seen.

Humans are also unreliable when selecting shades: studies have shown that only between 20% and 73% of shade selections of the same material under the same conditions by the same observer were identical on different days. Different observers also vary in their shade selections of the same shade of material over time.

Figure 15.11 shows a flowchart that should facilitate a more repeatable shade-matching routine; the information in parentheses briefly justifies each stage.

### Effects of colour vision deficiencies

While colour vision deficiency has been shown to have an adverse effect on ability to match dental colours, it also appears to have an unequal effect on the descriptors of colour. When compared to colour-normal subjects, CVD subjects have difficulty in recording hue and chroma but not value. It is difficult to define the effect of CVD on the complete shade matching process but is worthy of consideration if problems are evident. In addition, a CVD may affect the ability of individuals to differentiate restoration margins in an impression depending on the colour of the materials. This is especially true if two materials of different viscosity and colour are used.

## Instrumental shade matching

### Shade-matching devices

Instrumental shade-matching systems make use of colorimeters, spectrophotometers or digital images to make measurements at one or a number of points on a tooth's surface (Figure 15.12). The devices that use a spectrophotometer or colorimeter require a precise technique, keeping the probe at right angles to the tooth surface whilst keeping a steady hand (Figure 15.13). It is recommended that three readings are made and the most common used as the shade to be transferred to the laboratory. The recorded shade is available to the dentist, patient and technician almost immediately and most devices are able to provide CIE data.

Little evidence is available about the use of these devices in clinical practice; however, they have the potential to improve inter- and intra-clinician repeatability with regard to their shade selections over time and to select shade matches that are more acceptable to patients and dentists. They can also be used by the technician to verify the shade of the restoration that has been fabricated.

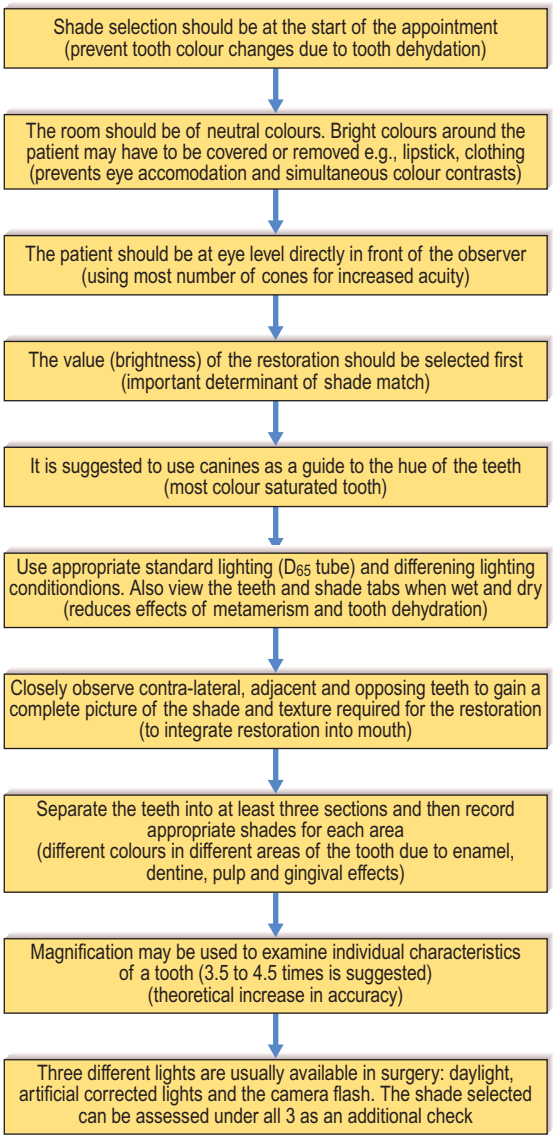



**Figure 15.11** Traditional shade matching flow chart.

### Digital photography

Information relating to shade and the aesthetic details discussed below can be transferred in a more objective manner by digital images. The shade cannot be calculated from a

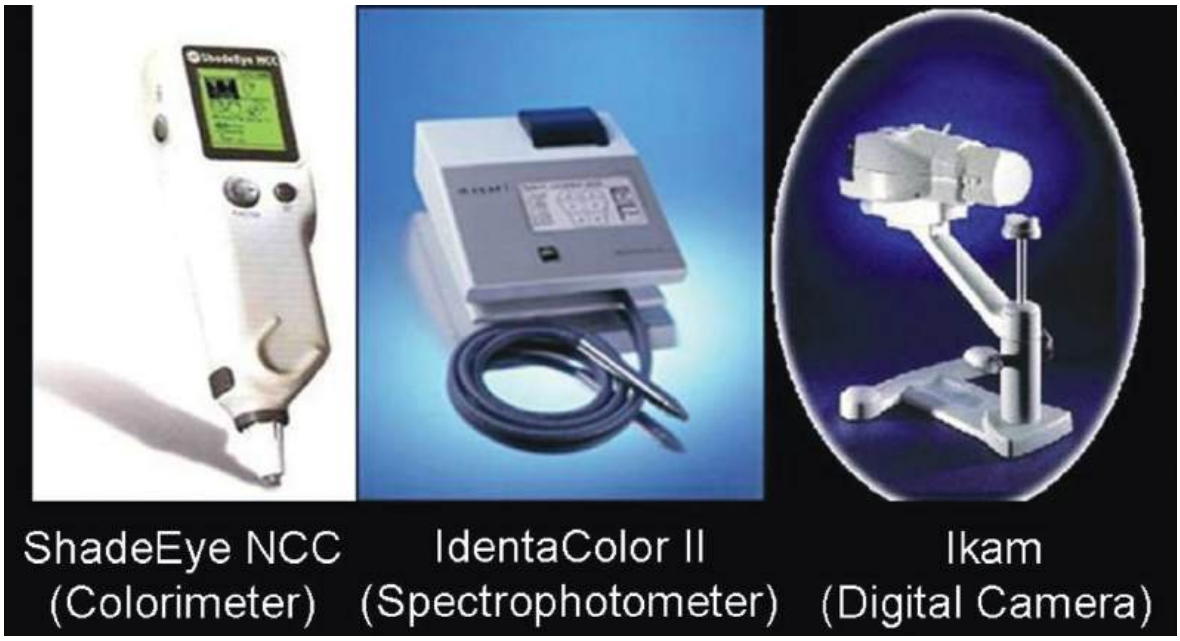



**Figure 15.12** Instrumental shade-matching devices.

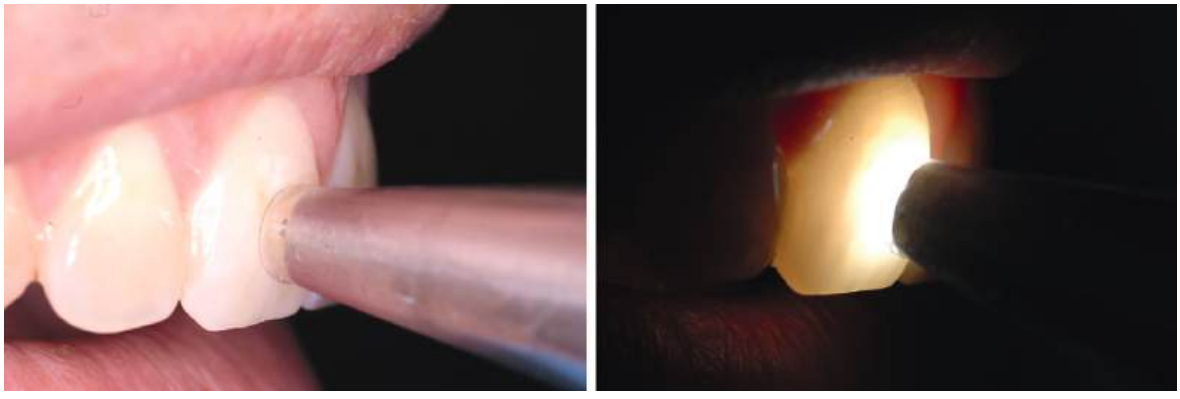


**Figure 15.13** Probe of instrumental shade-matching device, being approximated to the tooth at right angles (left) and the device recording the shade (right).

digital image without a calibration tile and conversion software. If digital images are to be used, then a series of images are usually required (Figure 15.14). In Figure 15.14 the camera was held at right angles to the tooth to give an accurate representation of translucency, effects, etc.

The minimum requirements for making intraoral images are a single-lens reflex (SLR) camera with an appropriate macro lens and flash. Two main types of flash are used: ring and a dual-point light. The former produces images that have no shadows as the buccal corridors are illuminated, producing good all-round images. A dual-point light allows shadows to assess texture and subtle changes in tooth surface anatomy. Compact cameras may be used but the images are often less accurate and the macro (close-up) lens is not consistently accurate enough for intraoral images. In addition, the flash often gives teeth a washed-out appearance and prevents the reproduction of subtle tooth characteristics. If digital images are used, a secure memory card is also required to store images in a manner that conforms to data protection regulations.



**Figure 15.14** Digital photographs to aid the technician. It is important that several views are taken. Note the shade tabs are in the same horizontal and vertical plane as the reference tooth and the shade is visible to the technician on at least one of the photographs.

## SUPPLEMENTAL AESTHETIC INFORMATION

Correctly matching the shade of a restoration to a tooth will not alone give an aesthetic result due to the complex structure of a tooth. Information relating to the subtleties of tooth form is also required. For example, vertical and horizontal tooth form need to be recorded, including specific characteristics such as transition lines, emergence profile, lustre, translucency, opalescence, dentine and enamel effects, stain lines, areas of hypomineralization, occlusal, cuspal/incisal and gingival colouration, root simulation, mammelons and lobe formation. In addition, conditions not related to tooth form have to be transferred to the laboratory and these should include tooth inclination (ideally progressive mesial angulation), gingival contours and levels, lip line and symmetry. Most of this type of information can be transferred via articulated study casts and diagnostic wax-ups during the diagnostic phase of treatment (see Chapter 5).

### Aesthetic analysis

The aesthetic demands and expectations of individual patients vary considerably and are often different from those of the treating clinician. Personnel from dental specialities might be more discerning than laypersons; however, some patients can be quite critical. Individuals from different dental specialities may also have different perceptions of aesthetics, with orthodontists being more critical of minor deviations in symmetry than prosthodontists for example. It is therefore important that realistic results are discussed with patients as well as the potential shortcomings of certain treatment modalities.

The aesthetic demands of people range from low to individuals where their perception of appearance is altered by psychological conditions (body dysmorphophobia). The aesthetic demands of some of these patient groups can be impossible to achieve and part of the initial assessment should attempt to identify such patients.

Excellent aesthetic integration can be achieved by placing full coverage restorations on all teeth in the upper anterior sextant; however, adverse biological outcomes have to be discussed with the patient (loss of pulp vitality, possible gingival recession, extensive treatment, maintenance and replacement, etc.). It is much harder, but not impossible, to achieve consistent aesthetic results when individual teeth are treated and more conservative treatments are used.

Before embarking on treatments that are going to have a high impact on a patient's appearance, it is important to consider the facial, dento-facial and dento-gingival relationships to complete an aesthetic analysis as well as managing the patients' expectations.

### Facial relationships

When viewed in the frontal plane, it has been stated that many face features are linked by orientation and size; however, rarely does one exhibit a perfectly symmetrical face.

#### *Facial midline*

Unfortunately, bilateral landmarks (eyes, ears, etc.) are often not exactly symmetrical: slight nasal deviations, small differences in the amount of eye opening and eye levels are found. This makes assessment of the facial midline difficult. Two reproducible points commonly used to identify a midline are nasion (midpoint between the two eyebrows) and the centre of the base of the philtrum. The direction of the midline can also be determined by these (Figure 15.15).

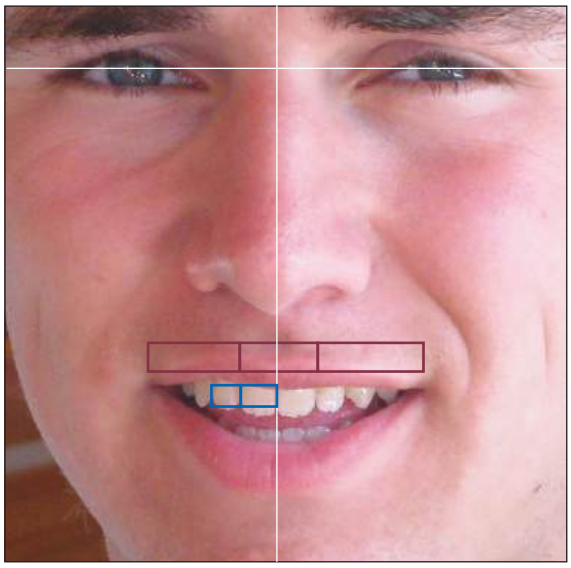


**Figure 15.15** Horizontal and vertical lines showing the interpupillary horizontal line and facial midline. The boxes show aspects of dental golden proportion.

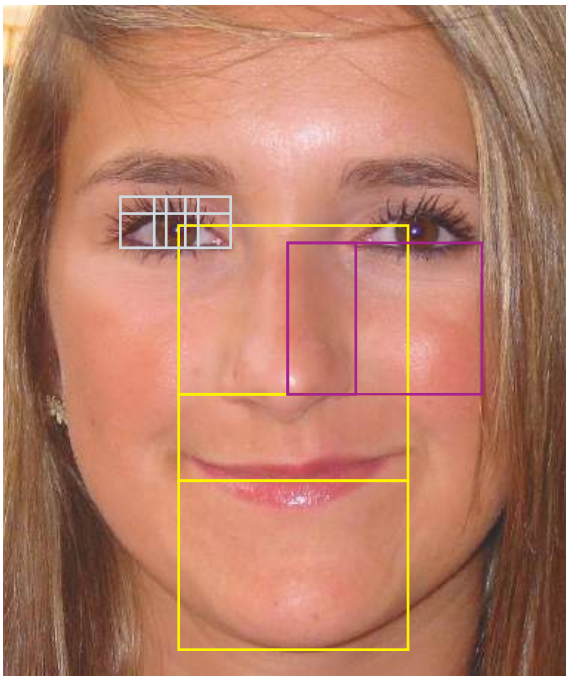


**Figure 15.16** Facial golden proportions.

### *Horizontal lines of the face*

It is often stated that the horizontal lines (interpupillary, commissural, nasal lines and occlusal plane) of the face should be parallel; however, this is often not the case. As the eyes are often at slightly different levels, it can be difficult to assess the relationship of the horizontal lines of the face using the interpupillary line as a guide. Therefore a better guide for the smile line is often to fabricate it at right angles to the facial midline (Figure 15.15).

### *Golden proportion*

The ancient Greeks discovered a complex ratio they named the 'golden mean' or 'phi', equalling 1.618, which was considered the most aesthetically pleasing ratio. The human body is said to have many excellent examples of phi, or golden proportion: the combined length of one's hand and forearm divided by the length of one's forearm is said to produce phi. Several ratios in the head and neck are purported to exist and may indicate an aesthetically pleasing face (Figures 15.15 and 15.16). The relationship of facial structures in the sagittal plane must also be considered. Skeletal, and dental, Class II and III relationships often prevent facial features from conforming to the golden proportion and are therefore considered by some to be less aesthetic.

In reality, it appears few individuals conform to all of the ideal ratios; however, it is known that aspects such as marked asymmetry, substantial deviation of facial and dental midlines, changes in face height and lateral discrepancies such as retrognathia can cause patients to be concerned about their appearance.

## Dento-facial relationships

### *Midline*

As discussed, the dental and facial midlines should be coincident or at least parallel to each other (Figure 15.15). The dental midline (line between upper and lower central incisors) is often examined relative to the midpoint of the upper lip (the middle of the middle lobe of the upper lip), but this may not be coincident with the facial midline. The dental and facial midlines are not coincident in approximately a quarter of patients. A deviation of the dental midline from the facial midline of 2.2 ± 1.5 mm has been found to be acceptable by various dental and lay observers, with about half of observers being unable identify differences between the facial and dental midlines if the discrepancy is less than 2 mm.

### *Smile line*

The relationship of the incisal edges of the maxillary anterior teeth and the lips can range from the parallel through to an inverted (reverse) relationship. When the upper central incisor and canine teeth rest on the lip curvature and the upper lateral incisors are above this line, the most pleasing outline is produced.

The smile line is assessed in several ways:

- The position of the upper lip to the maxillary gingival tissue (the most aesthetic smile is thought to exist when the full height of the maxillary teeth and the interproximal papillae are visible)
- The relationship of the curvature of the lower lip to the maxillary incisal tooth curvature
- The position of the maxillary incisal tooth curve relative to the lower lip.

These three factors can help determine the final contour and shape of restorations. It can be more difficult to treat patients with a high lip line as restoration contour, finish line placement and integration, gingival contour and colour are more critical. Assessing the lip line can be difficult as patients often produce a forced, unnatural smile. Asking a patient to say 'E' can help.

### *Vertical dimension*

Vertical dimension relates to the proportions of the lower face height compared to the upper face height. Facial aesthetics may be affected by the dental occlusal vertical dimension (OVD); however, substantial changes in OVD are required to alter the facial height significantly. The lower face height would be expected to be reduced in cases of extreme wear; however, due to compensatory growth of the dento-alveolar complexes this is often not the case.

## Dento-gingival relationships

### *The hard tissues*

Teeth come in a large variety of shapes and orientations relative to other teeth in the same mouth and between individuals (Figure 15.17). In general, teeth should have progressive mesial inclination (Figure 15.18) to give a pleasing appearance. The position and length of the interdental contacts are paramount to a pleasing smile: the contact between maxillary central incisors is the most coronal and the contacts move apically as one moves more posteriorly. The 50-40-30 rule applies to the length of these contacts. This rule relates to the relationship of the interdental contact height between the maxillary central incisors, the central incisor and lateral incisor, and the lateral incisor and canine teeth relative to the height of the central incisor. Fifty per cent of the length of the central incisors should ideally be involved in the contact zone, 40% for the contact between the lateral incisor and central incisor, with 30% for the canine and lateral incisor teeth.

Incisal openings between teeth should increase in size starting at the midline and moving posteriorly. Tooth form helps contribute to this as maxillary lateral incisors often have a more rounded mesial edge than central incisors and the shape of the upper canine (cuspid) lends itself to a larger embrasure.

It is generally accepted that black triangles or spacing between teeth is less aesthetic than teeth with dental and gingival interdental closure (Figure 15.19). The cause of increased interdental spacing is usually periodontal disease or a tooth:arch size discrepancy. This spacing can be reduced by having long interdental contacts, but the excessive contacts can lead to a poor appearance ('tombstoning').

Teeth are individual and exhibit substantial variation in tooth form (Figure 15.17). Disorders of tooth formation (hypomineralization, hypomaturation, amelogenesis imperfecta, dentinogenesis imperfecta) can further affect the appearance (see Chapter 5) and lead to demands for extensive restorative work.

#### *Tooth colour*

When considering colour and creating the optimum appearance it is important to bear in mind that teeth within the same mouth vary – for example, the colour of canine teeth are often darker and more yellow than the incisors (Figure 15.20). In addition, the relative amount of value, hue and chroma varies across each individual tooth (Figure 15.20). There is obviously a considerable variability in the colour of teeth between individuals and within individuals; the range of colours of restorative materials is vast but still do not cover the range of tooth colours observed. It is worth re-iterating that teeth naturally darken with age.

#### *Tooth size*

The ideal width:length ratio of an upper central incisor is 0.75:0.8. Shorter and wider teeth have a smaller ratio. These ratios provide guidelines to achieve good dental aesthetics and if the ratio is repeated throughout the maxillary sextant then the appearance can be excellent. This ratio can be assessed using a golden proportion ruler/gauge (Figure 15.21). This proportion must be calculated when the patient is viewed directly from the front.

The size of teeth varies both between individuals and over one's lifetime. Many studies have measured tooth dimensions and the dimensions determined by Magne et al. (2003) are shown in Table 15.1. This table also demonstrates the changes that occur due to tooth wear, with the length of teeth obviously getting shorter as they wear. This alters the width:length ratio which may adversely affect appearance.

#### *Bony factors*

The alveolar bone may influence the appearance of a smile. For example, excess alveolar contours (buccal bony tori) and dehiscences/fenestrations leading to soft tissue defects may be present.



Congenitally absent lateral incisors with comouflaged canine teeth



Absent upper right lateral incisor, retained deciduous canine, midline discrepancy and microdontic upper left lateral incisor providing a compromised aesthetic result



Microdontia of upper lateral incisors



Distal inclination of upper right central incisor and excessive mesial inclination of upper left incisors giving a compromised aesthetic result



Restorative interdental closure: long contact points and pink ceramic

**Figure 15.17** Variation in tooth form and appearance.

### *Soft tissues*

Poor gingival health and a history of periodontal disease can lead to gingival aesthetic problems. Gingival health must be gained and established as a prerequisite to aesthetic success. Gingival health must be established prior to placing definitive restorations. The position of the free gingival margin, muco-gingival junction and the width of the attached gingiva have to be assessed and corrected if required.



**Figure 15.18** Pleasing appearance of the upper anterior sextant.



**Figure 15.19** Poor contour of the central incisor implant retained crowns and loss of interdental papilla have led to unsightly black triangles between the restorations.

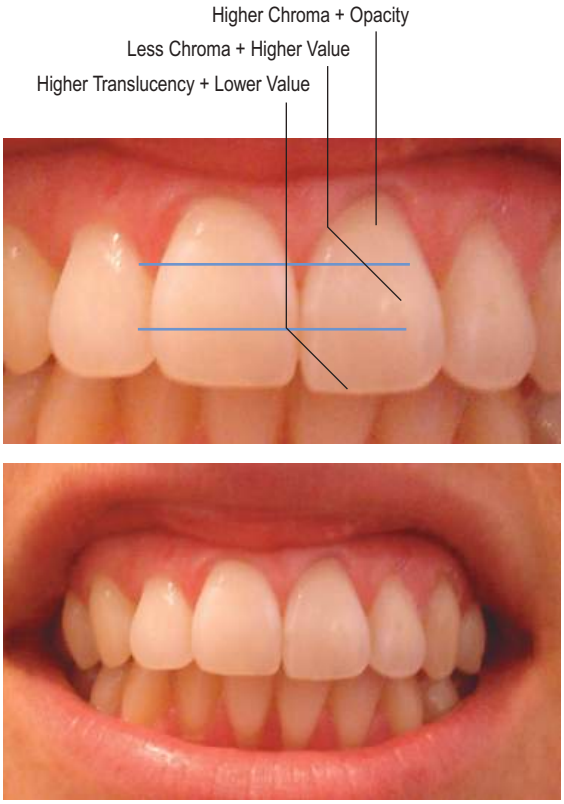



**Figure 15.20** Tooth colour variation across a tooth (above) and between teeth (below).

### *Gingival contour*

It is generally accepted that balance/symmetry and the correct proportions of the factors discussed are needed to create an aesthetic smile. In addition, the gingival contour is perhaps one of the most important factors in aesthetic dentistry. The example in Figure 15.22 shows that although the shade and size of restoration are acceptable, the gingival margins are not balanced. In addition to gingival health, complete gingival papillae, balanced gingival levels and correct gingival contours, colour and texture are required for a pleasing appearance (see Figure 15.18). It is generally accepted that the gingival margin should be more coronal around lateral incisors than central incisors or canines, and that the canine margin is the most apical of the maxillary anterior sextant. The gingival zenith should lie to the distal of each tooth's centre line (see Figure 15.18).

#### *Relationship between hard and soft tissues*

Periodontal characteristics have been studied in relation to crown shape. Two periodontal biotypes (thin and thick) have been identified which correlated to crown form (short-wide or long-narrow); patients with long-narrow teeth have been shown to be more prone to gingival recession than patients with short-wide teeth. This relationship has to be considered in patients with high aesthetic demands/high lip line as the risk of unsightly crown margins being exposed over time is increased.



**Figure 15.21** Golden proportion ruler.

**Table 15.1** Tooth dimensions

| | NO. OF TEETH MEASURED | WIDTH | LENGTH | W/L RATIO |
|---|---|---|---|---|
| Centrals, unworn (range) | 18 | **9.10** (0.62) (8.46–11.07) | **11.69** (0.70) (10.70–13.51) | **0.78** (0.03) (0.71–0.84) |
| Centrals, worn (range) | 26 | **9.24 (**0.66) (8.21–10.34) | **10.67** (1.13) (8.56–13.42) | **0.87** (0.08) (0.74–1.10) |
| Laterals, unworn (range) | 30 | **7.07** (0.76) (5.51–8.22) | **9.75** (0.83) (8.19–11.51) | **0.73** (0.07) (0.57–0.83) |
| Laterals, worn (range) | 11 | **7.38** (0.52) (6.43–7.89) | **9.34** (0.80) (7.97–11.22) | **0.79** (0.06) (0.70–0.87) |
| Canines, unworn (range) | 25 | **7.90** (0.64) (6.80–9.02) | **10.83** (0.77) (9.71–12.94) | **0.73** (0.06) (0.60–0.82) |
| Canines, worn (range) | 13 | **8.06** (0.74) (6.60–8.72) | **9.90** (0.84) (8.83–11.77) | **0.81** (0.06) (0.72–0.91) |
| Premolars (range) | 23 | **7.84** (0.73) (6.61–8.84) | **9.33** (0.94) (7.66–10.45) | **0.84** (0.06) (0.65–0.95) |

From Magne et al. (2003).



**Figure 15.22** The aesthetic result for this patient is compromised by the unbalanced gingival margins, despite acceptable shade matches of the metal–ceramic crown on the upper left central incisor and the minimum preparation (resin retained) bridge replacing the upper right lateral incisor.

## SUMMARY

Creating the best aesthetics in indirect restorations is a complex process and involves far more than simply colour matching. This chapter has looked in depth at shade matching as well as all associated features of tooth form and how this is framed within the supporting periodontal tissues, the smile line and facial features. Having an understanding of these will enable the dentist to achieve the highest aesthetics possible within a given situation, assess appearance critically and be able to inform patients about the limitations of possible treatment options.

## FURTHER READING

Magne, P., Gallucci, G.O., Belser, U.C., 2003. Anatomic crown width/length ratios of unworn and worn maxillary teeth in white subjects. J. Prosthet. Dent. 89, 453–461.

Miller, A., Long, J., Cole, J., Staffanou, R., 1993. Shade selection and laboratory communication. Quintessence Int. 24, 305–309.

Wasson, W., Schuman, N., 1992. Color vision and dentistry. Quintessence Int. 23, 349–353.